Lecture 2:Business Analytics & Machine Learning Basics

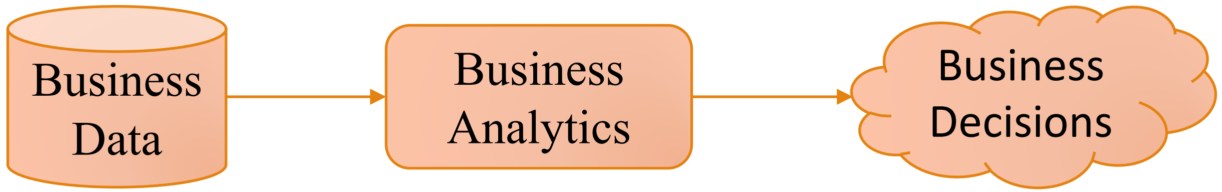
Overview

- Business Analytics

- Machine learning basics
  - Classification
  - Numeric prediction
  - Model evaluation

Business Analytics

- **Business Analytics (BA)** refers to the skills, technologies, practices for continuous iterative exploration and investigation of past business performance to gain insight and drive business planning.



Business Analytics

- **Business Analytics** requires quantitative methods and evidence-based data for business modeling and decision making.
  - Large volume of data
    - Structured data
    - Unstructured
  - When companies analyze data, they are using Business Analytics to get the insights required for making better business decisions and strategic moves.

Different Types of Data

- Different types of data

Class Roster

Course: MGT 500 Business Policy

Semester: Spring 2010

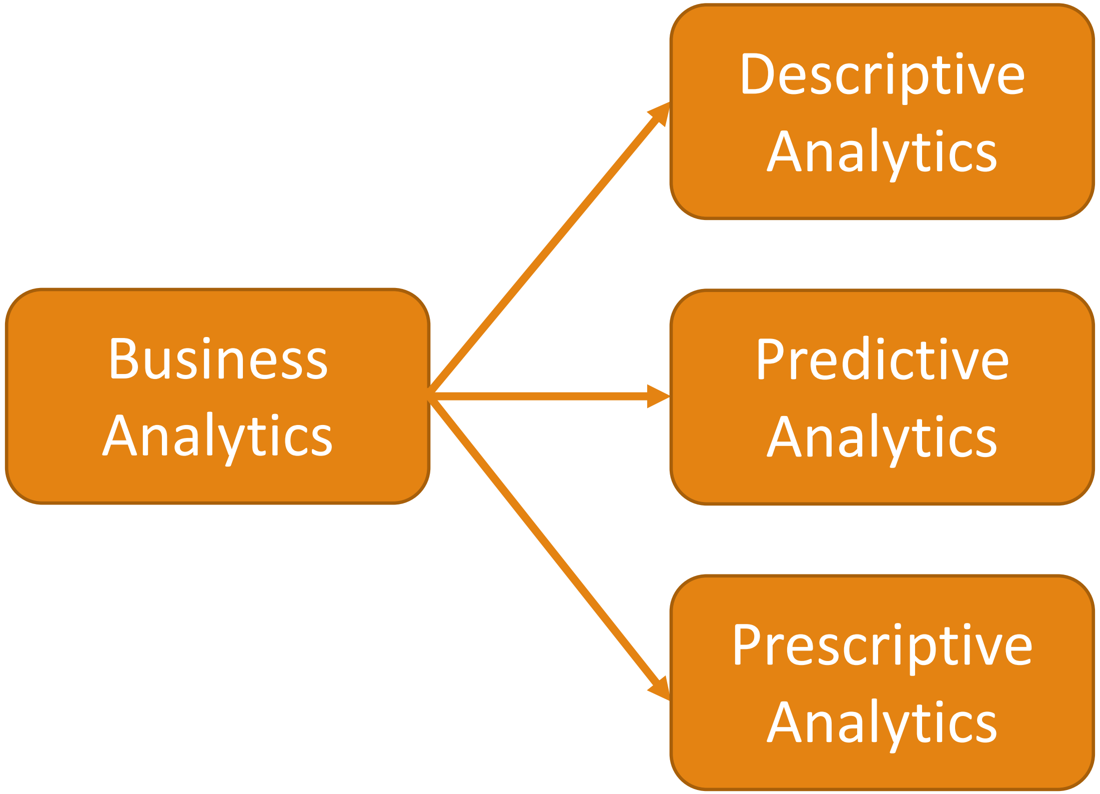
Section: 2

Name	ID	Major	GPA
Baker, Kenneth D.	324917628	MGT	2.9
Doyle, Joan E.	476193248	MKT	3.4
Finkle, Clive R.	548429344	PRM	2.8
Lewis, John C.	551742186	MGT	3.7
McFerran, Debra R.	409723145	IS	2.9
Sisneros, Michael	392416582	ACCT	3.3

Structured data: data stored in tabular form

Unstructured data: image, video, audio, text

Business Analytics Techniques



Business Analytics Techniques

- **Descriptive** Analytics, which use data aggregation and data mining to provide insight into the past and answer: "What has happened?"
- **Predictive** Analytics, which use statistical models and forecasts techniques to understand the future and answer: "What could happen?"
- **Prescriptive** Analytics, which use optimization and simulation algorithms to advice on possible outcomes and answer: "What do we need to do to achieve this?"

Business Analytics – Predictive Analytics

- **Predictive Analytics**
- Encompasses a variety of statistical techniques from predictive modelling, machine learning, and data mining that analyze current and historical facts to make predictions about future or otherwise unknown events.
- In business, predictive models exploit patterns found in historical and transactional data to identify risks and opportunities. Models capture relationships among many factors to allow assessment of risk or potential associated with a particular set of conditions, guiding decision making for candidate transactions.

Business Analytics – Predictive Analytics

- **Machine Learning**: Machine learning is a branch of artificial intelligence based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention.

  - Classification
  - Numeric Prediction
  - Clustering
  - Association Rule Mining

Predictive Analytics– Business Application

- **Fraud detection**

- ***PayPal*** uses data such as your historical payment data, to the kind of device often used as well as your PayPal user profile and country of origin, all these go into building machine learning algorithms that detect potential signs of fraud with every transaction.

Predictive Analytics– Business Application

- **Order predicting**

- ***Amazon*** uses data analysis to predict what buyers are going to buy before they buy it. Using data such as historical buying patterns, demographics, browsing habits, and wish lists, Amazon can predict what customers are going to purchase. To take advantage of this, the company has patented "Anticipatory Shipping." Utilizing their customer data and forecasting spikes and lulls in demand, Amazon moves items to warehouses near the individuals who are most likely to buy those products. The benefit of this is decreased shipping time, with the eventual goal of delivering an item to a customer within a few hours.

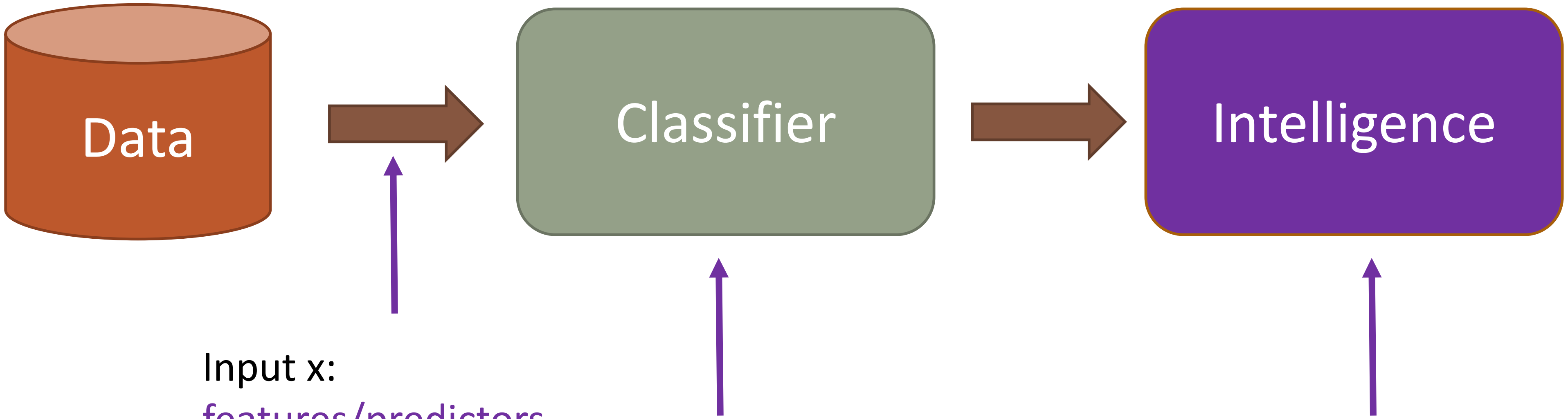
Machine Learning Basics - Classification

What is Classification?

- Classify objects into a set of **pre-specified** classes (or categories) based on the values of relevant object variables (features).

What is Classification?

From features to predictions



Input x: features/predictors derived from data

Learn x-> y relationships

Predict y: categorical "output"

Kicked Vehicle

- One of the biggest challenges of an auto dealership purchasing a used car at an auto auction is the risk of that the vehicle might have serious issues that prevent it from being sold to customers. The auto community calls these unfortunate purchases "kicks".
- Kicked cars often result when there are tampered odometers, mechanical issues the dealer is not able to address, issues with getting the vehicle title from the seller, or some other unforeseen problem. Kick cars can be very costly to dealers after transportation cost, throw-away repair work, and market losses in reselling the vehicle.

Kicked Vehicle

- Figure out which cars have a higher risk of being kick can provide real value to dealerships trying to provide the best inventory selection possible to their customers.

- Problem: Predict if the car purchased at the Auction is a Kick (bad buy).

What we know about a car?

- Auction: Auction provider at which the vehicle was purchased
- Color: Vehicle Color
- MMRCurrentAuctionAveragePrice: Acquisition price for this vehicle in average condition as of current day
- Size: The size category of the vehicle (Compact, SUV, etc.)
- TopThreeAmericanName: Identifies if the manufacturer is one of the top three American manufacturers
- VehBCost: Acquisition cost paid for the vehicle at time of purchase
- VehicleAge: The Years elapsed since the manufacturer's year
- VehOdo: The vehicles odometer reading
- WarrantyCost: Warranty price (term=36month and millage=36K)
- WheelType: The vehicle wheel type description (Alloy, Covers)

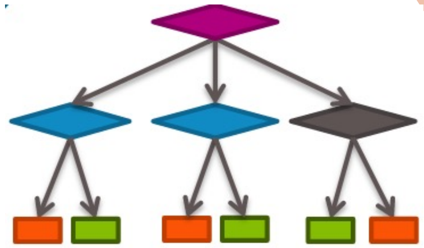
- What don’t we know about a car?

What we know about a car?

Information of Cars

- Auction
- Color
- MMRCurrentAuctionAveragePrice
- Size
- TopThreeAmericanName
- VehBCost
- VehicleAge
- VehOdo
- WarrantyCost
- WheelType

Predictors (X)



Classifier

IsBadBuy

Target variable /Label (Y)

Predictor and Target Variable

- **Predictor**: A predictor (or **predictor variable**) is a variable whose values will be used to predict the value of the target variable.
- **Target variable**: The target variable (or **outcome variable**) is the variable whose values are to be modeled and predicted by other variables.
- **Classifier**: an algorithm/classification model that implements classification (mapping input data to a category).

Data

Auction	Color	IsBadBuy	MMRCurrent	Size	TopThreeAm	VehBCost	VehicleAge	VehOdo	WarrantyCos	WheelType
ADESA	WHITE	No	2871	LARGE TRUC	FORD	5300	8	75419	869	Alloy
ADESA	GOLD	Yes	1840	VAN	FORD	3600	8	82944	2322	Alloy
ADESA	RED	No	8931	SMALL SUV	CHRYSLER	7500	4	57338	588	Alloy
ADESA	GOLD	No	8320	CROSSOVER	FORD	8500	5	55909	1169	Alloy
ADESA	GREY	No	11520	LARGE TRUC	FORD	10100	5	86702	853	Alloy
ADESA	SILVER	No	2659	COMPACT	GM	4100	7	73810	1455	Covers
ADESA	RED	No	4645	VAN	FORD	5600	5	85003	1633	Covers
ADESA	SILVER	No	4352	LARGE	GM	5900	5	88991	2152	Covers
ADESA	SILVER	No	5142	MEDIUM	GM	6600	5	80077	1373	Alloy
ADESA	MAROON	No	9983	MEDIUM	OTHER	7500	3	71952	1272	Alloy
ADESA	WHITE	No	4165	MEDIUM	OTHER	6200	4	23881	462	Covers

Steps to Build A Classifier

- Step 1: Kicked Vehicle Data
- Step 2: Data Preprocessing
- Step 3: Build a decision tree on training data (learn the relationship between predictors and target variable)
- Step 4: Evaluate Decision Tree Model Performance

Step 1: Kicked Vehicle Data

	Auction	Color	IsBadBuy	MMRCurrentAuctionAveragePrice	Size	TopThreeAmericanName	VehBCost	VehicleAge	VehOdo	WarrantyCost	WheelType
0	ADESA	WHITE	No	2871	LARGE TRUCK	FORD	5300	8	75419	869	Alloy
1	ADESA	GOLD	Yes	1840	VAN	FORD	3600	8	82944	2322	Alloy
2	ADESA	RED	No	8931	SMALL SUV	CHRYSLER	7500	4	57338	588	Alloy
3	ADESA	GOLD	No	8320	CROSSOVER	FORD	8500	5	55909	1169	Alloy
4	ADESA	GREY	No	11520	LARGE TRUCK	FORD	10100	5	86702	853	Alloy
...	...	...	...	...	...	...	...	...	...	...	...
9995	ADESA	RED	No	7536	SMALL SUV	CHRYSLER	6600	4	85377	983	Alloy
9996	ADESA	BLACK	No	4921	LARGE TRUCK	GM	7000	7	89665	1543	Alloy
9997	ADESA	BLACK	No	9263	MEDIUM SUV	CHRYSLER	9000	4	59383	1417	Alloy
9998	ADESA	BLUE	No	3240	MEDIUM	OTHER	5500	4	48642	482	Covers
9999	ADESA	SILVER	No	6148	MEDIUM SUV	FORD	7000	5	82563	1243	Alloy

10000 rows × 11 columns

Target variable

Steps to build a decision tree model

- Step 1: Kicked Vehicle Data
- Step 2: Data Preprocessing
- Step 3: Build a decision tree on training data (learn the relationship between predictors and target variable)
- Step 4: Evaluate Decision Tree Model Performance

Step 2: Data Preprocessing

- **Dummy coding**
  - Dummy coding: convert categorical variables into numeric format
    - Create **dummy variables** to replace the original categorical variable
    - Dummy coding for a binary variable: gender

$$\text{male} = \begin{cases} 1 & \text{if x} = \text{male} \\ 0 & \text{otherwise} \end{cases}$$

Auction provider: ADESA, MANHEIM, OTHER

	Auction
Vehicle 1	OTHER
Vehicle 2	MANHEIM
Vehicle 3	ADESA

n categories, *n-1* dummy variables

	Auction_MANHEIM	Auction_OTHER
Vehicle 1	0	1
Vehicle 2	1	0
Vehicle 3	0	0

Step 2: Data Preprocessing

Target variable

	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	VehOdo	WarrantyCost	Auction_MANHEIM	Auction_OTHER	Color_BLACK	...	WheelType_unkwnWheel
0	No	2871	5300	8	75419	869	0	0	0		0
1	Yes	1840	3600	8	82944	2322	0	0	0		0
2	No	8931	7500	4	57338	588	0	0	0		0
3	No	8320	8500	5	55909	1169	0	0	0		0
4	No	11520	10100	5	86702	853	0	0	0		0
...	...	...	...	...	...	...	...	...	...	...	...
9995	No	7536	6600	4	85377	983	0	0	0		0
9996	No	4921	7000	7	89665	1543	0	0	1		0
9997	No	9263	9000	4	59383	1417	0	0	1		0
9998	No	3240	5500	4	48642	482	0	0	0		0
9999	No	6148	7000	5	82563	1243	0	0	0		0

10000 rows × 40 columns

Step 2: Data Preprocessing

Split the data to training and testing sets

	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	VehOdo	WarrantyCost	Auction_MANHEIM	Auction_OTHER	Color_BLACK	...	WheelType_unkwnWheel
0	No	2871	5300	8	75419	869	0	0	0		0
1	Yes	1840	3600	8	82944	2322	0	0	0		0
2	No	8931	7500	4	57338	588	0	0	0		0
3	No	8320	8500	5	55909	1169	0	0	0		0
4	No	11520	10100	5	86702	853	0	0	0		0
...	...	...	...	...	...	...	...	...	...	...	...
6995	No	2762	4185	4	84379	1243	0	1	0		0
6996	Yes	5144	8705	5	80600	1373	0	1	0		0
6997	No	2912	4485	6	77686	1389	0	1	0		0
6998	No	4965	6495	4	66731	1020	0	1	0		0
6999	No	2724	3985	6	86275	1243	0	1	0		0
...	...	...	...	...	...	...	...	...	...	...	...
9995	No	7536	6600	4	85377	983	0	0	0		0
9996	No	4921	7000	7	89665	1543	0	0	1		0
9997	No	9263	9000	4	59383	1417	0	0	1		0
9998	No	3240	5500	4	48642	482	0	0	0		0
9999	No	6148	7000	5	82563	1243	0	0	0		0

Training data

Testing data

Step 2: Data Preprocessing

- Holdout Evaluation
  - Splitting method (percentage split)- divide into training and testing sets (e.g. 70%/30% or 2/3 to 1/3)
    - 2 partitions, e.g.
      - 70% and 30% in training and testing
      - **Training and testing data are not overlapped**
      - Most commonly used to get a sense of a model's performance level

Steps to build a decision tree model

- Step 1: Kicked Vehicle Data
- Step 2: Data Preprocessing
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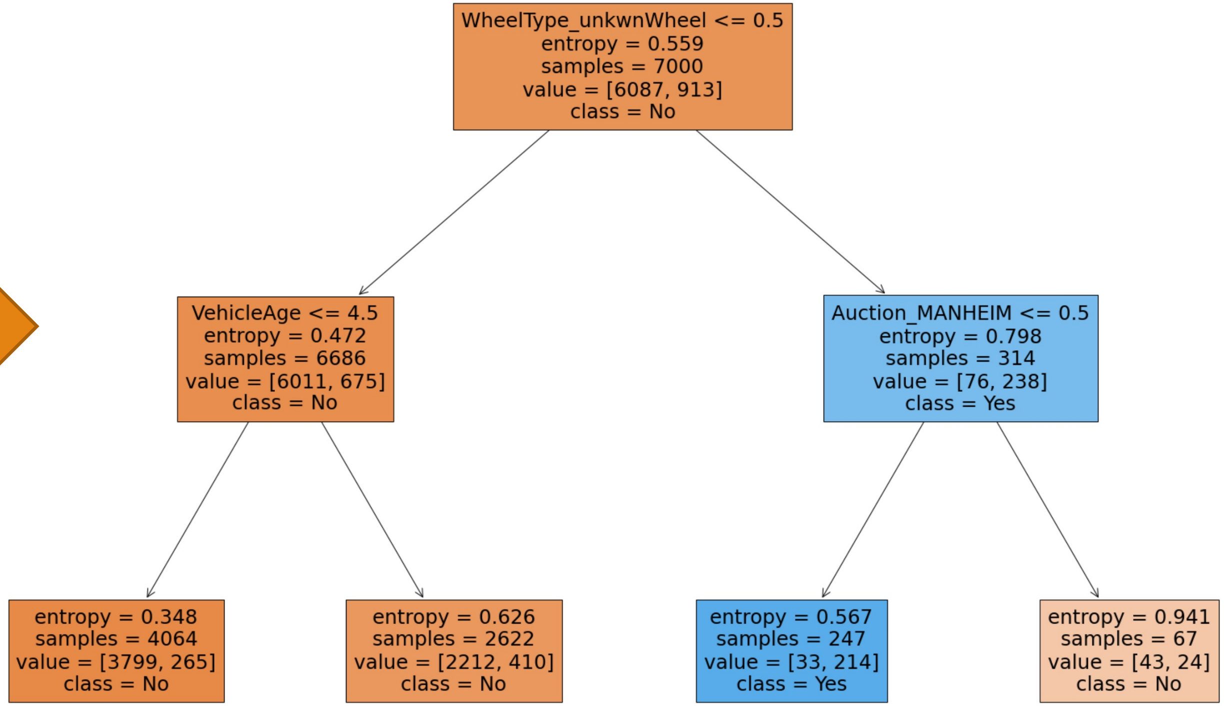
- Step 4: Evaluate Decision Tree Model Performance

Step 3: Build a decision tree on training data

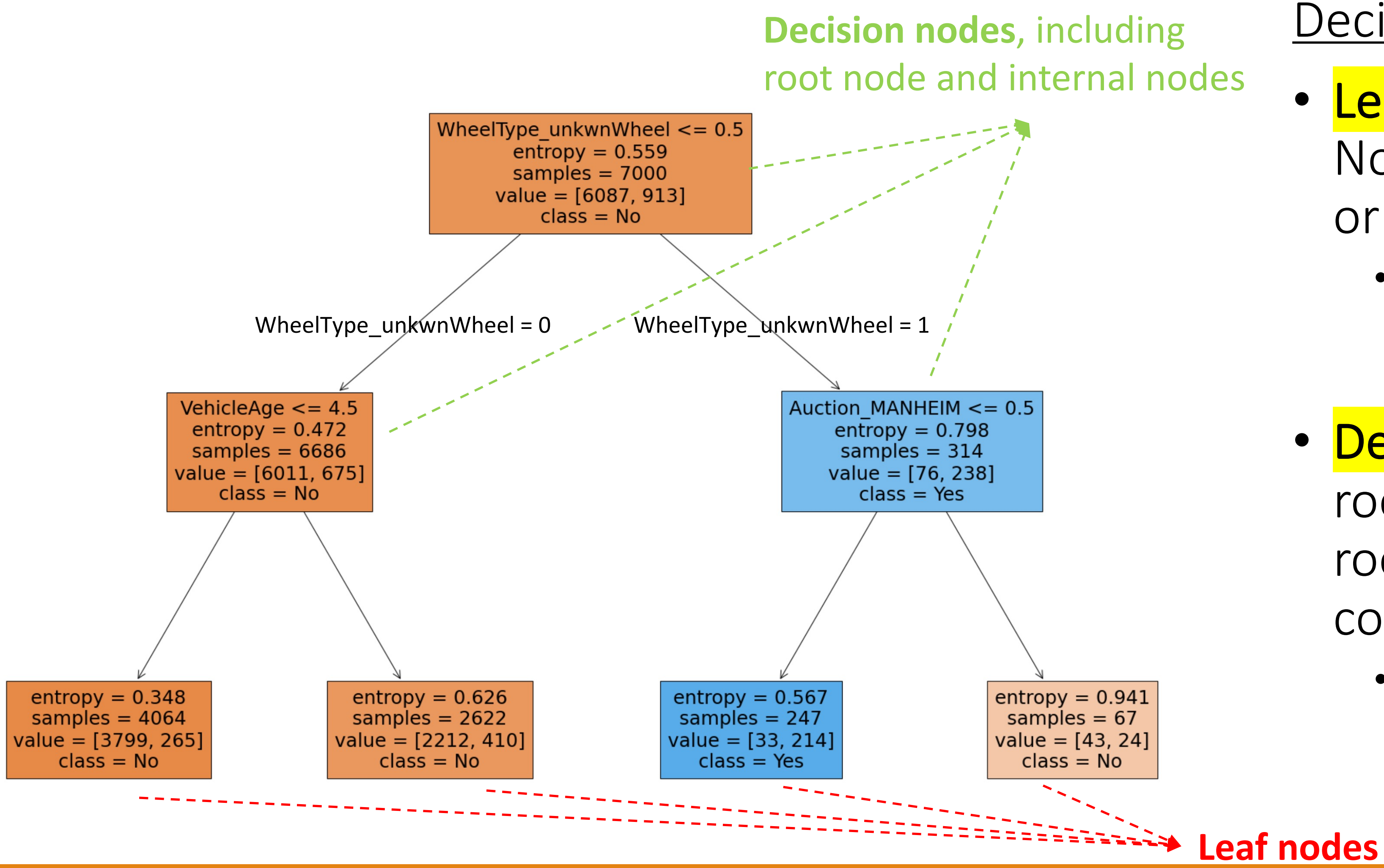
	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	…
0	No	2871	5300	8	
1	Yes	1840	3600	8	
2	No	8931	7500	4	
3	No	8320	8500	5	
4	No	11520	10100	5	
...	...	...	...	...	…
6995	No	2762	4185	4	
6996	Yes	5144	8705	5	
6997	No	2912	4485	6	
6998	No	4965	6495	4	
6999	No	2724	3985	6	

Training data



Decision Tree Model

Decision Tree

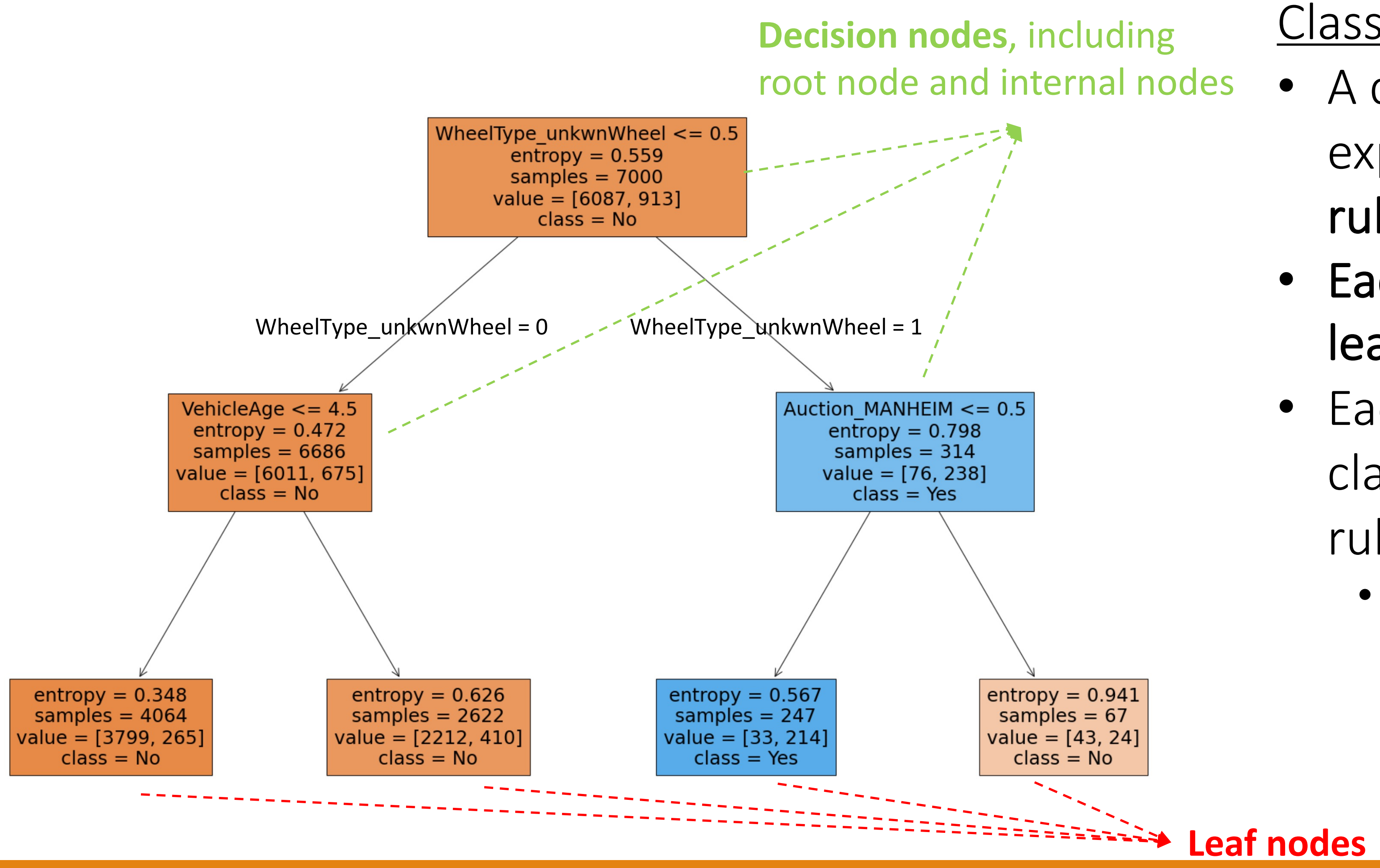


Decision Tree structure

- **Leaf Node**: The end of the tree. Nodes do not split is called Leaf or Terminal node.
  - A leaf node holds a **class label** (**outcome**): prediction result - Yes or No

- **Decision Node:** Includes both root node and internal node. The root node or an internal node contains a **predictor.**
  - **Root Node**: The first node of the tree. It represents entire population or sample and this further gets divided into two or more homogeneous sets.

Decision Tree

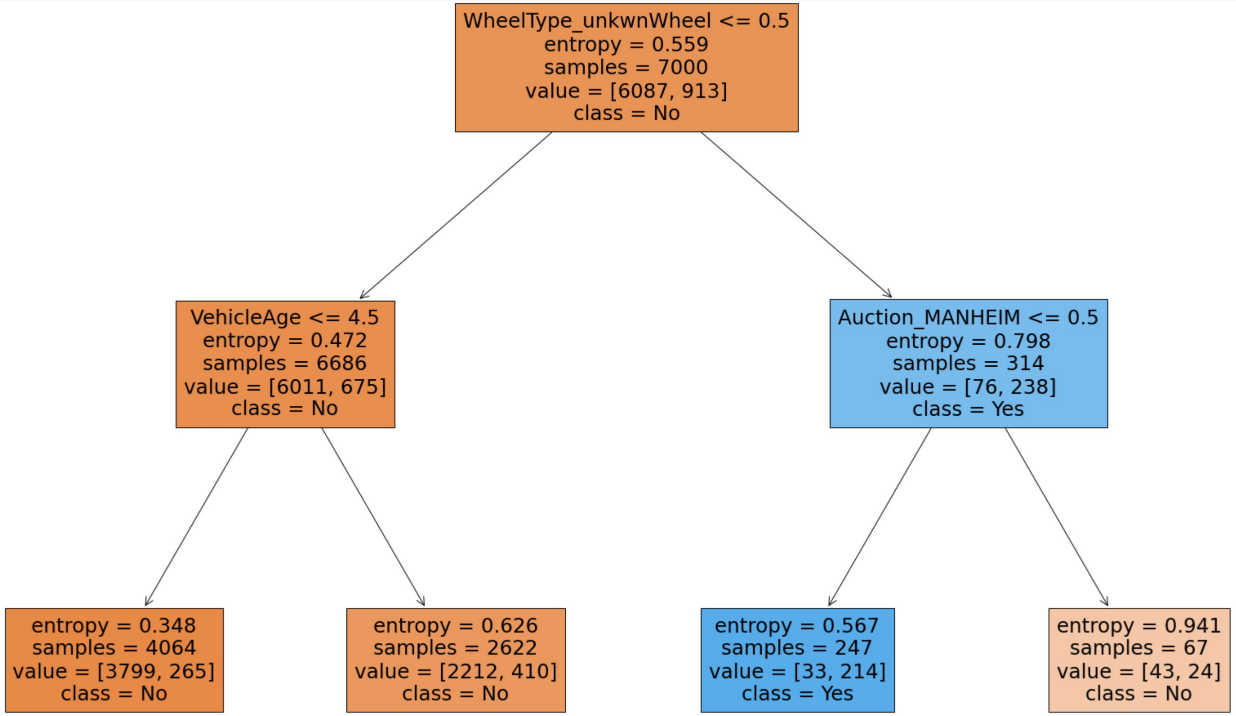


<u>Classification rules</u>

- A decision tree can be expressed as a set of **IF-THEN rules**.
- **Each path from the root to a leaf** forms an IF-THEN rule.
- Each observation/data record is classified based on an IF-THEN rule
  - **IF** WheelType = unkwnWheel, Auction = {ADESA, OTHER}, **THEN** IsBadBuy=YES

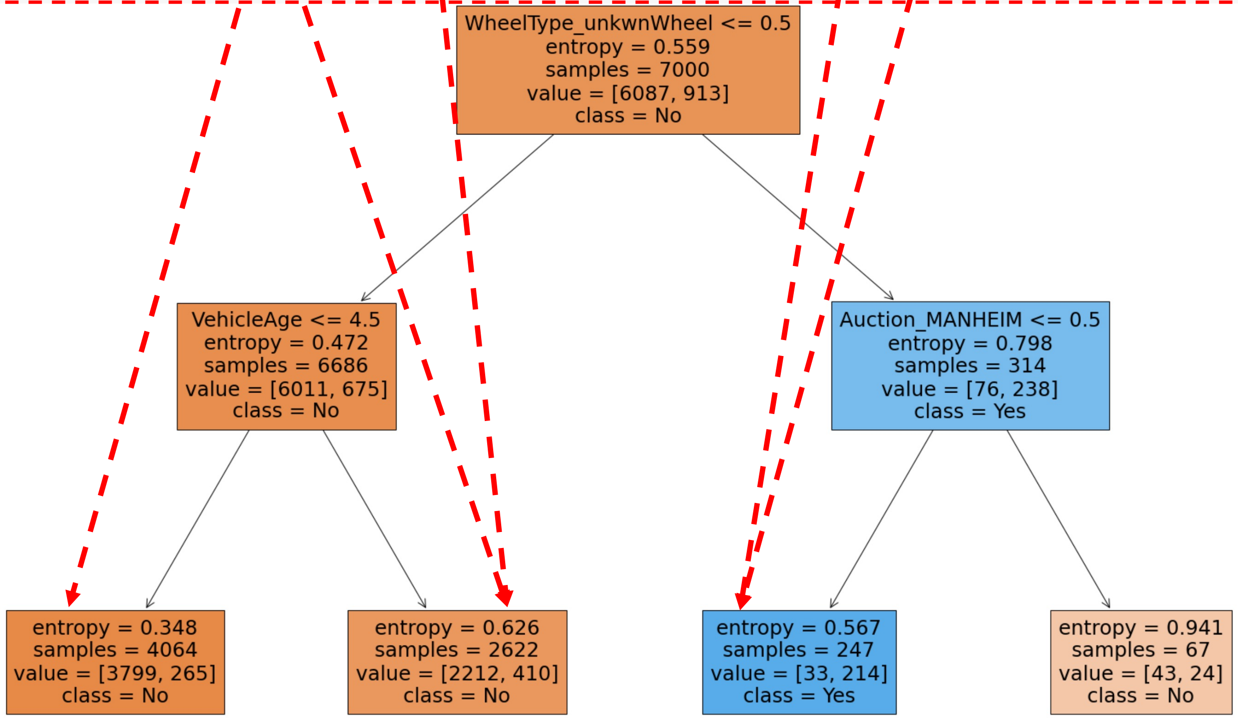
Decision Tree

	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	VehOdo	WarrantyCost	Auction_MANHEIM	Auction_OTHER	...	WheelType_Covers	WheelType_Special	WheelType_unkwnWheel
0		2871	5300	8	75419	869	0	0		0	0	0
1		1840	3600	8	82944	2322	0	0		0	0	1
2		8931	7500	4	57338	588	0	0		0	0	0
3		8320	8500	5	55909	1169	0	0	...	0	0	1
4		11520	10100	5	86702	853	0	0		0	0	0



Decision Tree

	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	VehOdo	WarrantyCost	Auction_MANHEIM	Auction_OTHER	...	WheelType_Covers	WheelType_Special	WheelType_unkwnWheel
0		2871	5300	8	75419	869	0	0		0	0	0
1		1840	3600	8	82944	2322	0	0		0	0	1
2		8931	7500	4	57338	588	0	0		0	0	0
3		8320	8500	5	55909	1169	0	0	...	0	0	1
4		11520	10100	5	86702	853	0	0		0	0	0



Steps to build a decision tree model

- Step 1: Kicked Vehicle Data
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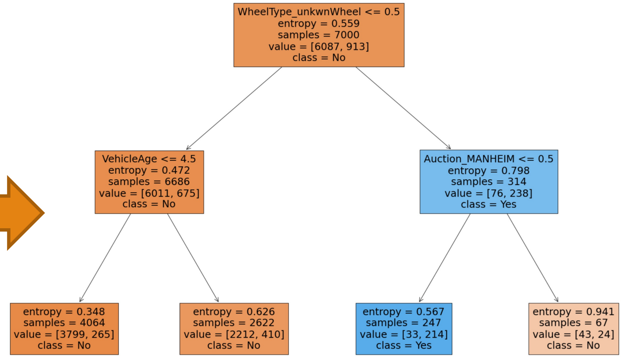
- Step 4: Evaluate Decision Tree Model Performance

Step 4: Evaluate Decision Tree Model Performance

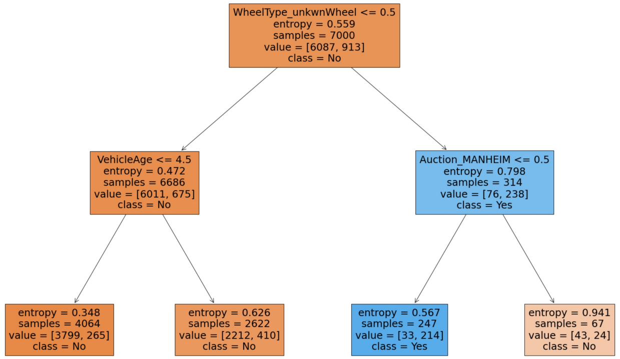
	IsBadBuy	MMRCurrentAuctionAveragePrice	VehBCost	VehicleAge	VehOdo	WarrantyCost	Auction_MANHEIM	Auction_OTHER	Color_BLACK
0	No	2871	5300	8	75419	869	0	0	0
1	Yes	1840	3600	8	82944	2322	0	0	0
2	No	8931	7500	4	57338	588	0	0	0
3	No	8320	8500	5	55909	1169	0	0	0
4	No	11520	10100	5	86702	853	0	0	0
...	...	...	...	...	...	...	...	...	...
6995	No	2762	4185	4	84379	1243	0	1	0
6996	Yes	5144	8705	5	80600	1373	0	1	0
6997	No	2912	4485	6	77686	1389	0	1	0
6998	No	4965	6495	4	66731	1020	0	1	0
6999	No	2724	3985	6	86275	1243	0	1	0
...	...	...	...	...	...	...	...	...	...
9995		7536	6600	4	85377	983	0	0	0
9996		4921	7000	7	89665	1543	0	0	1
9997		9263	9000	4	59383	1417	0	0	1
9998		3240	5500	4	48642	482	0	0	0
9999		6148	7000	5	82563	1243	0	0	0



Training data

testing data

Step 4: Evaluate Decision Tree Model Performance



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9996		4921	7000	7	89665	1543	0	0	1
9997		9263	9000	4	59383	1417	0	0	1
9998		3240	5500	4	48642	482	0	0	0
9999		6148	7000	5	82563	1243	0	0	0

testing data

Predictions

IsBadBuy
No
No
No
No
Yes
Yes

Real value

IsBadBuy
No
No
No
No
No
No

Step 4: Evaluate Decision Tree Model Performance

- Compare the **predictions** and **real values/actual value**

Predictions/predicted values

IsBadBuy
No
No
No
No
Yes
Yes

real values

IsBadBuy
No
No
No
No
No
No

Step 4: Evaluate Decision Tree Model Performance

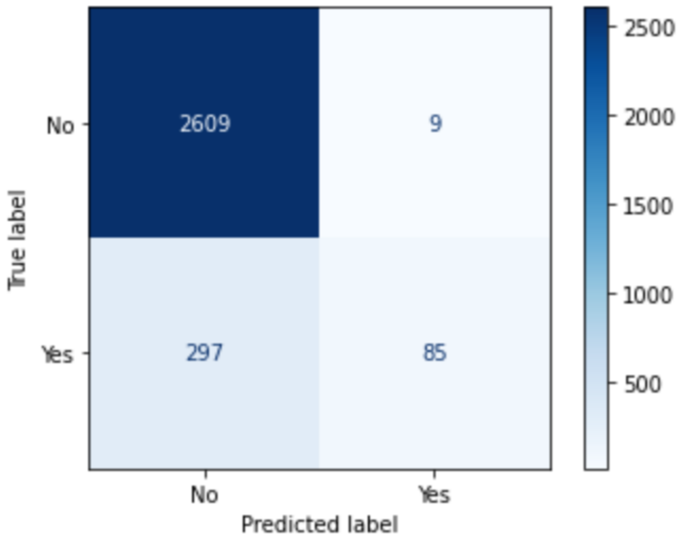
- Confusion Matrix
  - A **confusion matrix** is a table that categorizes predictions according to whether they match the actual value.

Step 4: Evaluate Decision Tree Model Performance

	Predicted Class Label		
True Class Label		**a**	**b**
	a	True Positive (TP)	False Negative (FN) (Type II error)
	b	False Positive (FP) (Type I error)	True Negative (TN)

- **True Positive (TP)**: Correctly classified as is positive class
- **True Negative (TN)**: Correctly classified as negative class
- **False Positive (FP)**: Incorrectly classified as positive class
- **False Negative (FN)**: Incorrectly classified as negative class

Assume **a** is positive class and **b** is negative class



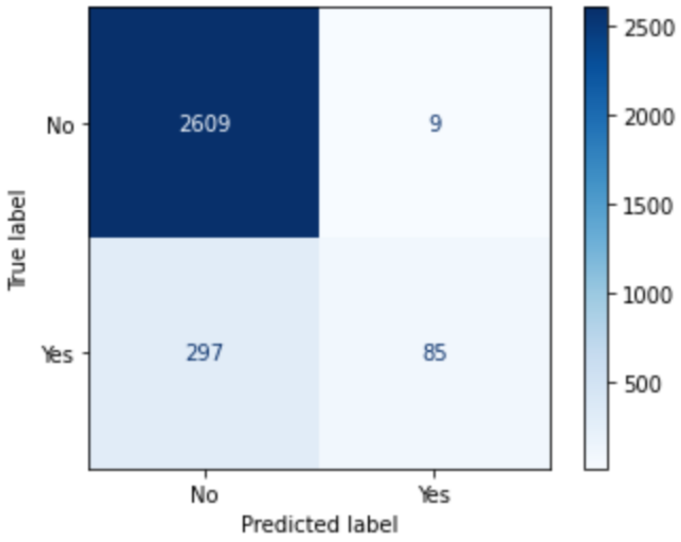
Step 4: Evaluate Decision Tree Model Performance

	Predicted Class Label		
True Class Label		**a**	**b**
	a	True Positive (TP)	False Negative (FN) (Type II error)
	b	False Positive (FP) (Type I error)	True Negative (TN)

- **a** is positive class
- **b** is negative class
- T (Total population) = TP+TN+FP+FN

- True class label is **a**= TP+FN
- Predicted class label is **a** = TP+FP
- True class label is **b** = FP+TN
- Predicted class label is **b** = FN+TN

Assume **a** is positive class and **b** is negative class



Step 4: Evaluate Decision Tree Model Performance

- **Accuracy** is the overall correctness of the model and is calculated as the sum of correct predictions divided by the total number of predictions.

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- The **error rate** or the proportion of the incorrectly classified examples is specified as

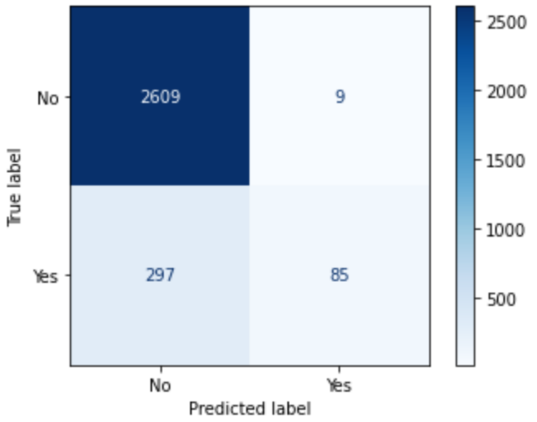
$$\text{error rate} = \frac{\text{FP} + \text{FN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = 1 - \text{accuracy}$$

Step 4: Evaluate Decision Tree Model Performance

- Evaluate performances on positive/negative class
  - **Precision**
  - **recall**
  - **F-measure**

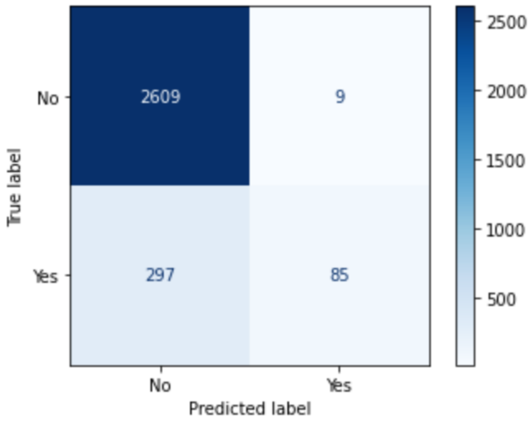
Step 4: Evaluate Decision Tree Model Performance

- Evaluation Metrics: Precision
  - The **precision** is defined as the proportion of predicted positive examples that are truly positive.
    - How many instances are predicted to the positive (**a**) class?
      - Predicted class label is **a** = TP+FP
    - How many of instances predicted as **a** actually belong to **a** class?
      - TP
  - Precision (**a** or positive) = TP/(TP+FP)=2609/(2609+297)
  - Precision (**b** or negative) = TN/(TN+FN) = 85/(85+9)



Step 4: Evaluate Decision Tree Model Performance

- Evaluation Metrics: Recall
  - **Recall** is the ability to correctly classify instances belonging to this class.
    - How many instances actually belong to the positive (**a**) class?
      - True class label is **a**= TP+FN
    - How many of instances predicted as **a** actually belong to **a** class?
      - TP
  - Recall (a or positive) = TP/(TP+FN) = 2609/(2609 + 9)
  - Recall (b or negative) = TN/(TN+FP) = 85/(85 + 297)

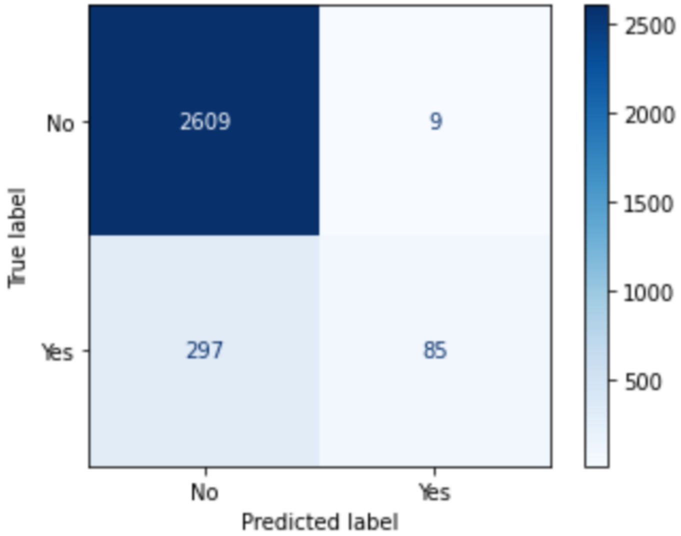


Step 4: Evaluate Decision Tree Model Performance

- F-measure
  - The harmonic mean of precision and recall.
  - It can be used as a single measure of overall performance on positive/negative class.
    - F-measure (**a**) = (2 x Precision(**a**) x Recall(**a**)) / (Precision(**a**) + Recall(**a**))
    - F-measure (**b**) = (2 x Precision(**b**) x Recall(**b**)) / (Precision(**b**) + Recall(**b**))

Step 4: Evaluate Decision Tree Model Performance

Performance on the **testing data**:



	precision	recall	f1-score	support
No	0.90	1.00	0.94	2618
Yes	0.90	0.22	0.36	382
accuracy			0.90	3000
macro avg	0.90	0.61	0.65	3000
weighted avg	0.90	0.90	0.87	3000

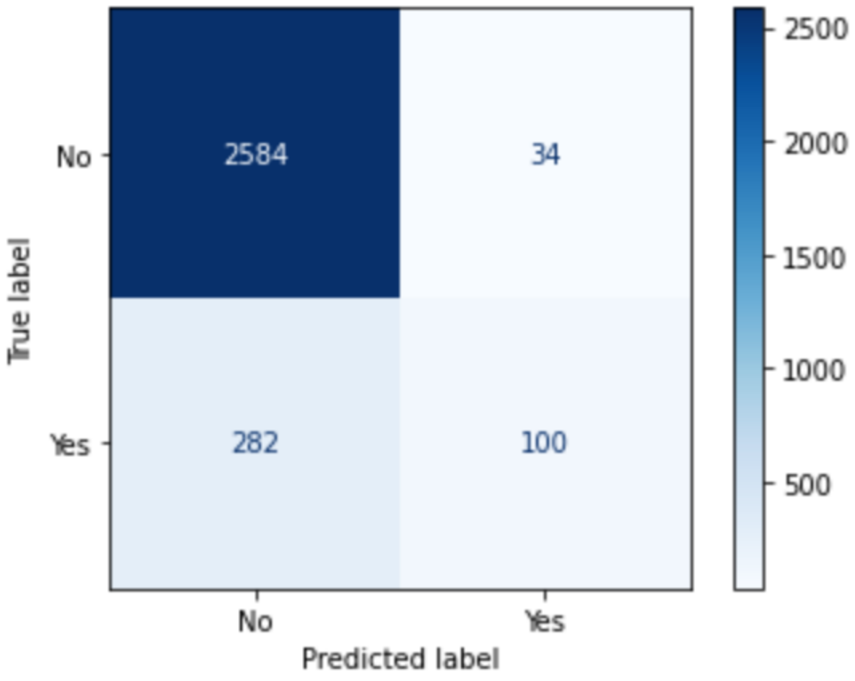
Overall performance on Not bad buy class

Overall performance on bad buy class

Model's overall performance

Step 4: Evaluate Model Performance

Performance on the **testing data**:



	precision	recall	f1-score	support
No	0.90	0.99	0.94	2618
Yes	0.75	0.26	0.39	382
accuracy			0.89	3000
macro avg	0.82	0.62	0.66	3000
weighted avg	0.88	0.89	0.87	3000

1. How many cars are predicted as Not bad buy?
2. How many bad buy cars are predicted as Not bad buy?
3. How many bad buy cars are identified correctly by the model?
4. If the model predicts a car as bad buy, what is the probability that such prediction is correct?
5. Does the model have better performance on majority (IsBadBuy = 'No') or minority class (IsBadBuy = 'Yes')?

Model Comparison

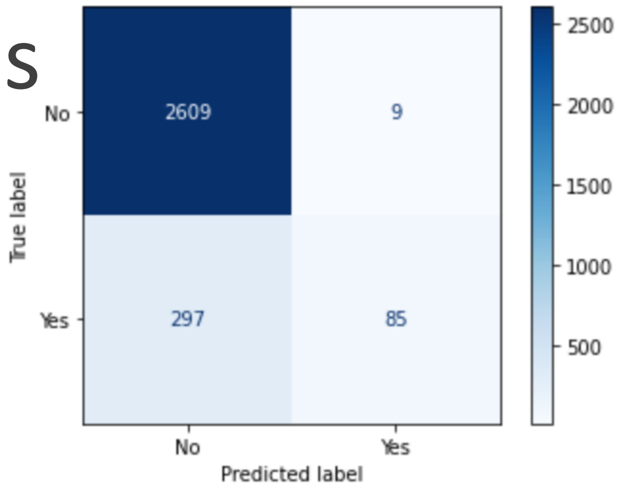
- Comparing Decision Tree and Naïve Bayes
  - Decision Tree

	precision	recall	f1-score	support
No	0.90	1.00	0.94	2618
Yes	0.90	0.22	0.36	382
accuracy			0.90	3000
macro avg	0.90	0.61	0.65	3000
weighted avg	0.90	0.90	0.87	3000

Overall performance on NOT bad buy class

Overall performance on bad buy class

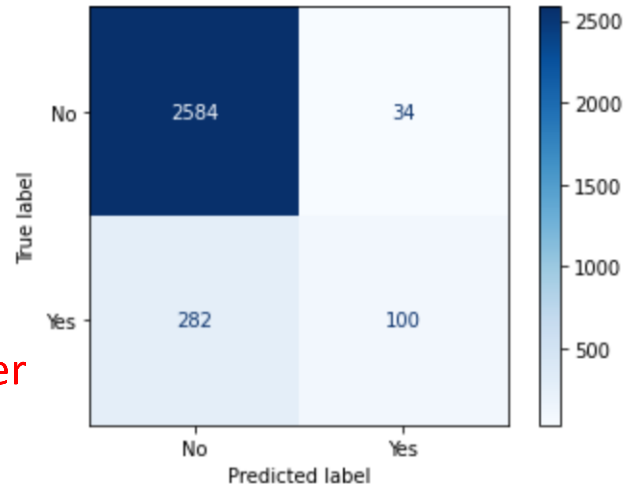
Model's overall performance



  - Naïve Bayes

	precision	recall	f1-score	support
No	0.90	0.99	0.94	2618
Yes	0.75	0.26	0.39	382
accuracy			0.89	3000
macro avg	0.82	0.62	0.66	3000
weighted avg	0.88	0.89	0.87	3000

1. Which model has better performance?
2. Which model is better to identify bad cars?



Model Comparison

- Overall model performance comparison
  - Accuracy
- Compare performances on each class
  - F-measure: single metrics combines precision and recall and measures the overall performance on each class
  - Precision: confidence/effectiveness of predictions
  - Recall: ability of identifying instances belonging to a class

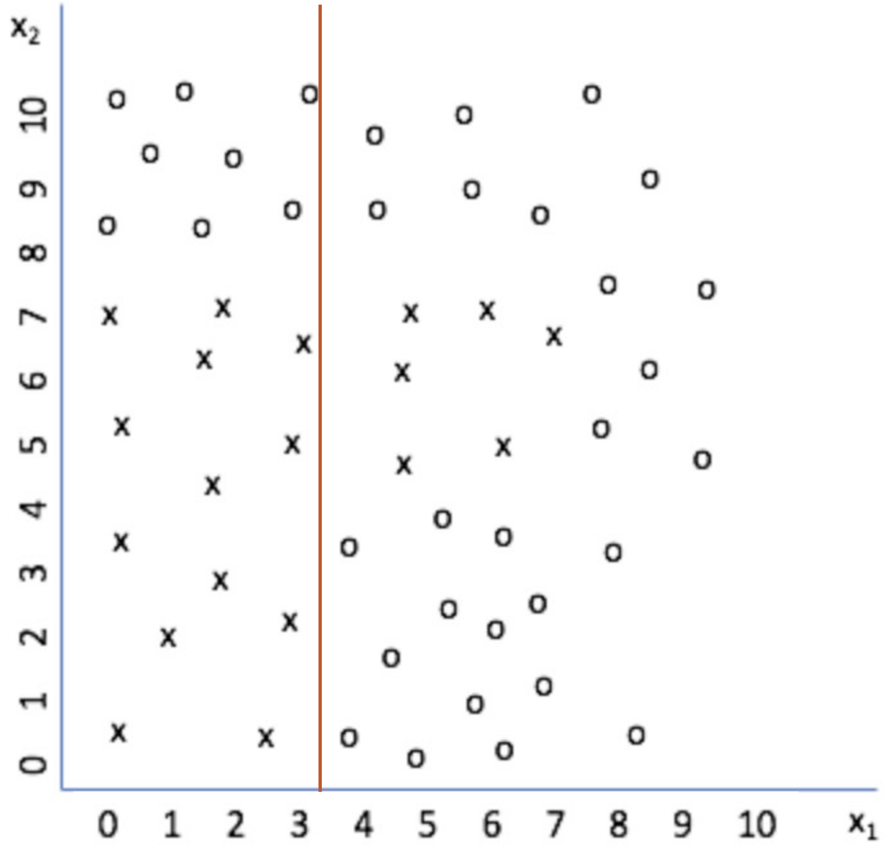
Steps to build a decision tree model

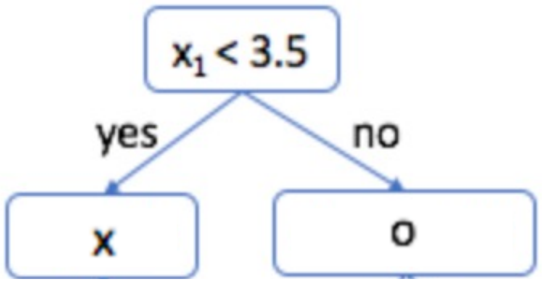
- Step 1: Kicked Vehicle Data
- Step 2: Data Preprocessing
- Step 3: Build a decision tree on training data (learn the relationship between predictors and target variable)
- Step 4: Evaluate Decision Tree Model Performance

Machine Learning Basics – Generalization and Overfitting

Generalization and Overfitting

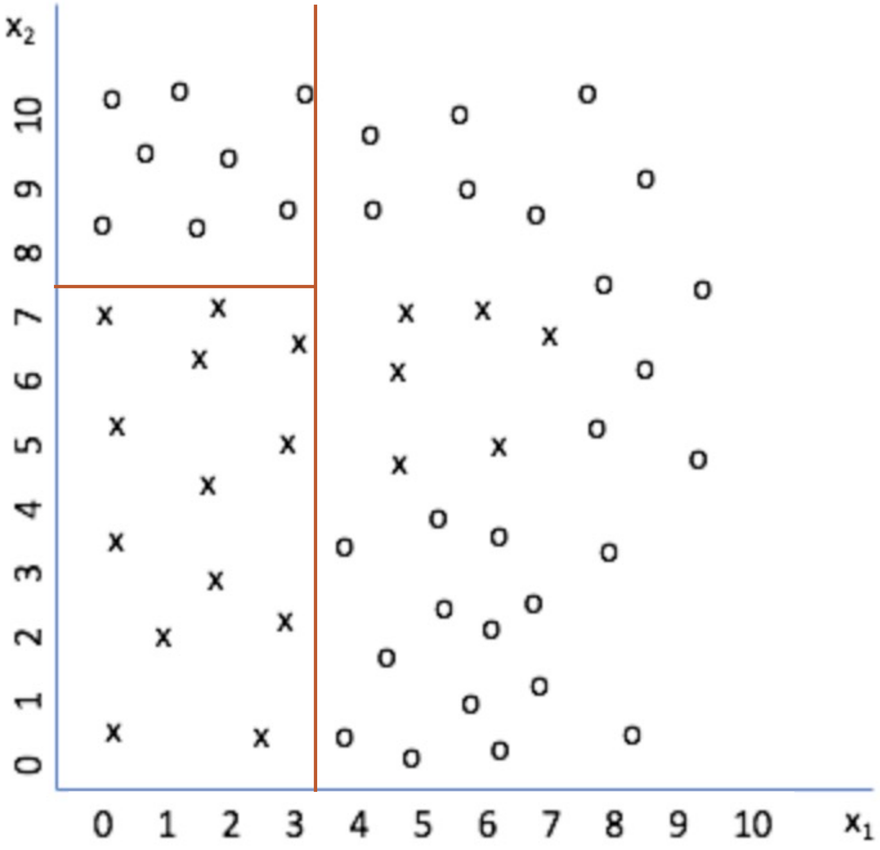
- Decision Tree

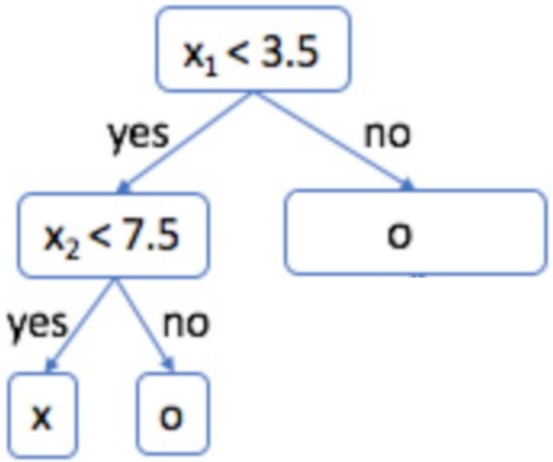




Generalization and Overfitting

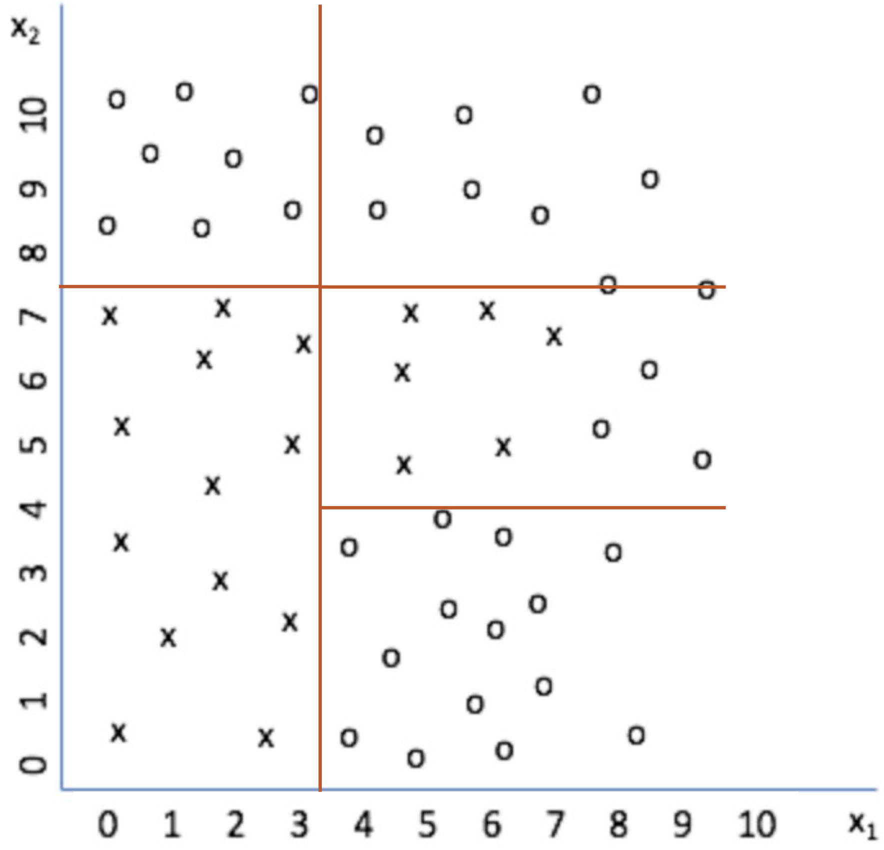
- Decision Tree

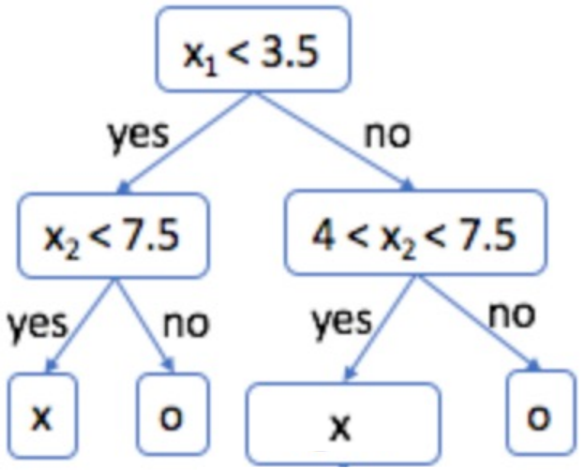




Generalization and Overfitting

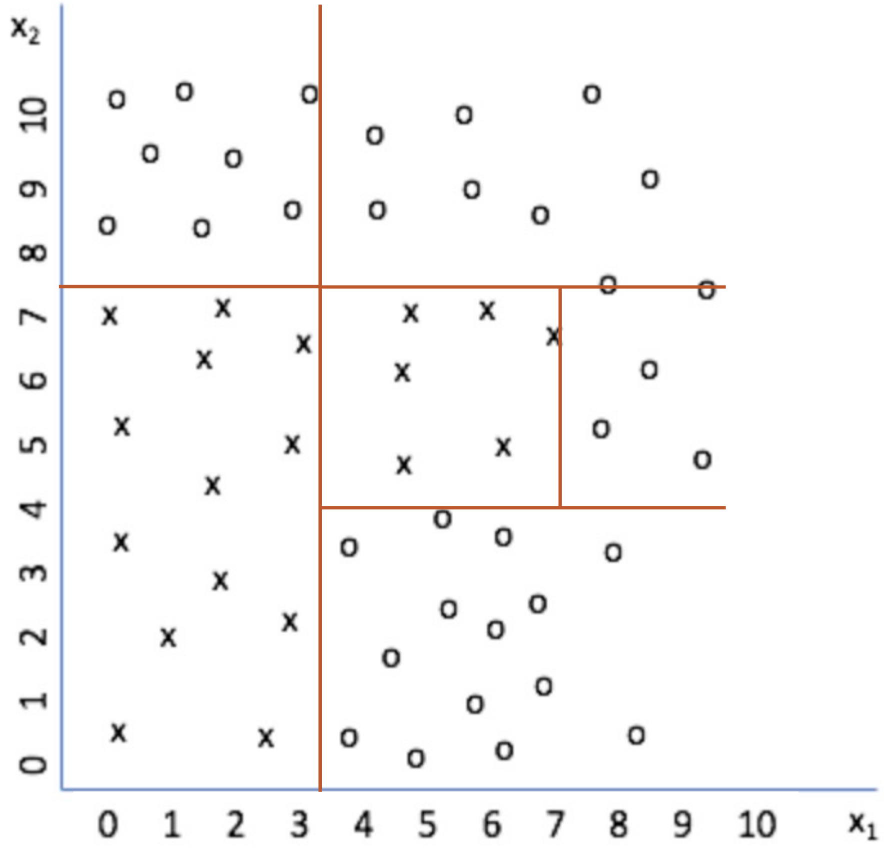
- Decision Tree

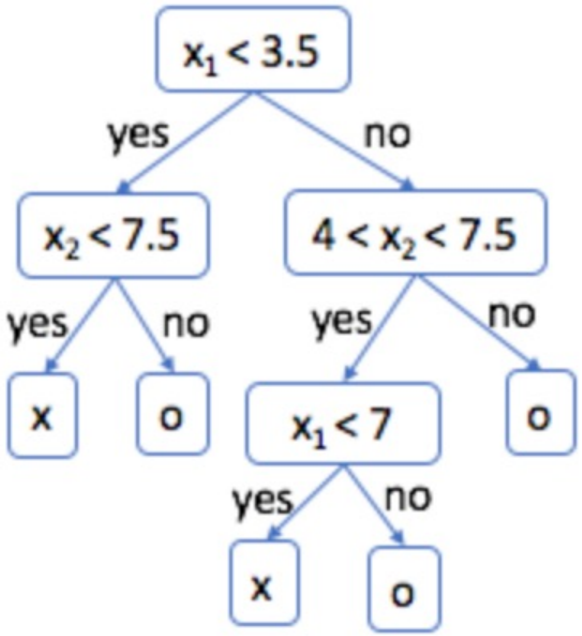




Generalization and Overfitting

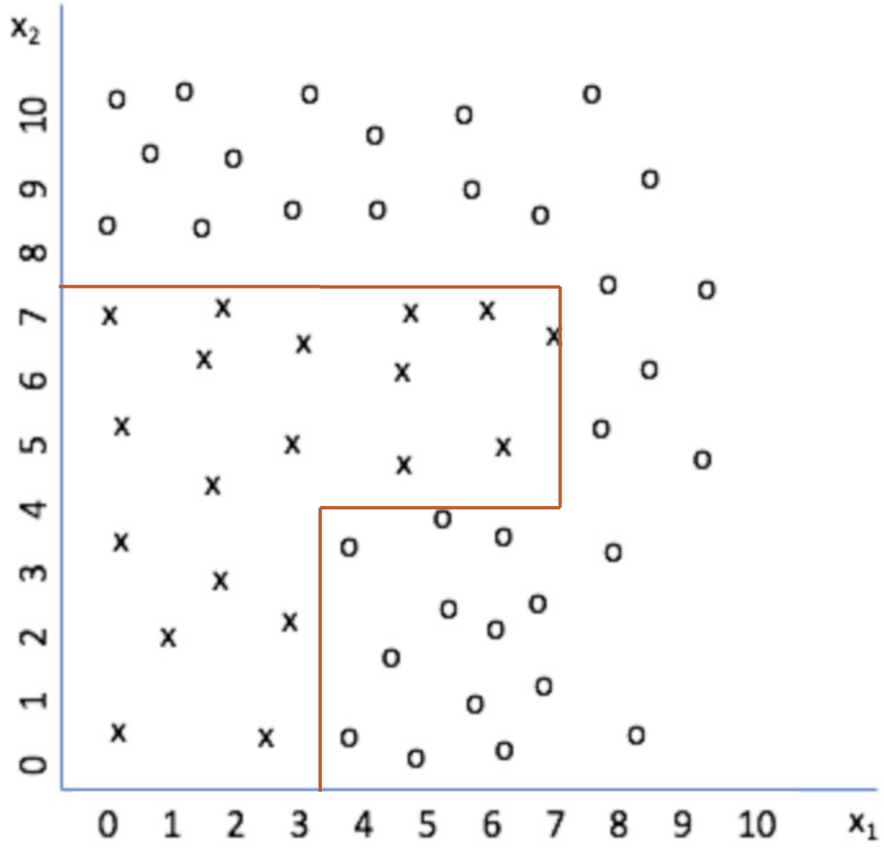
- Decision Tree

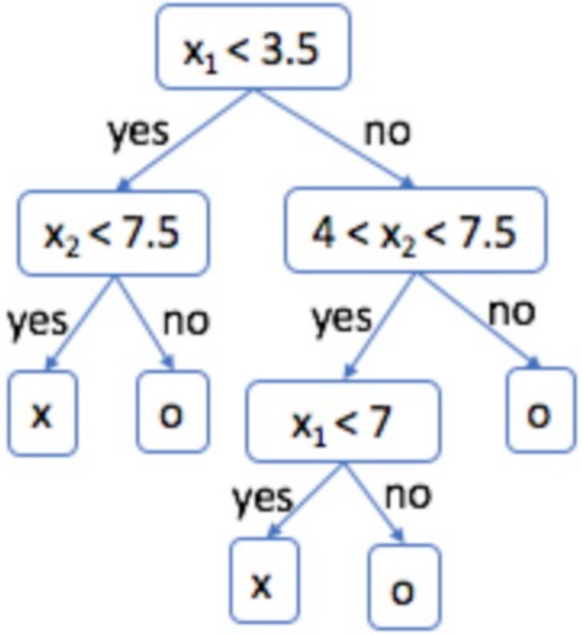




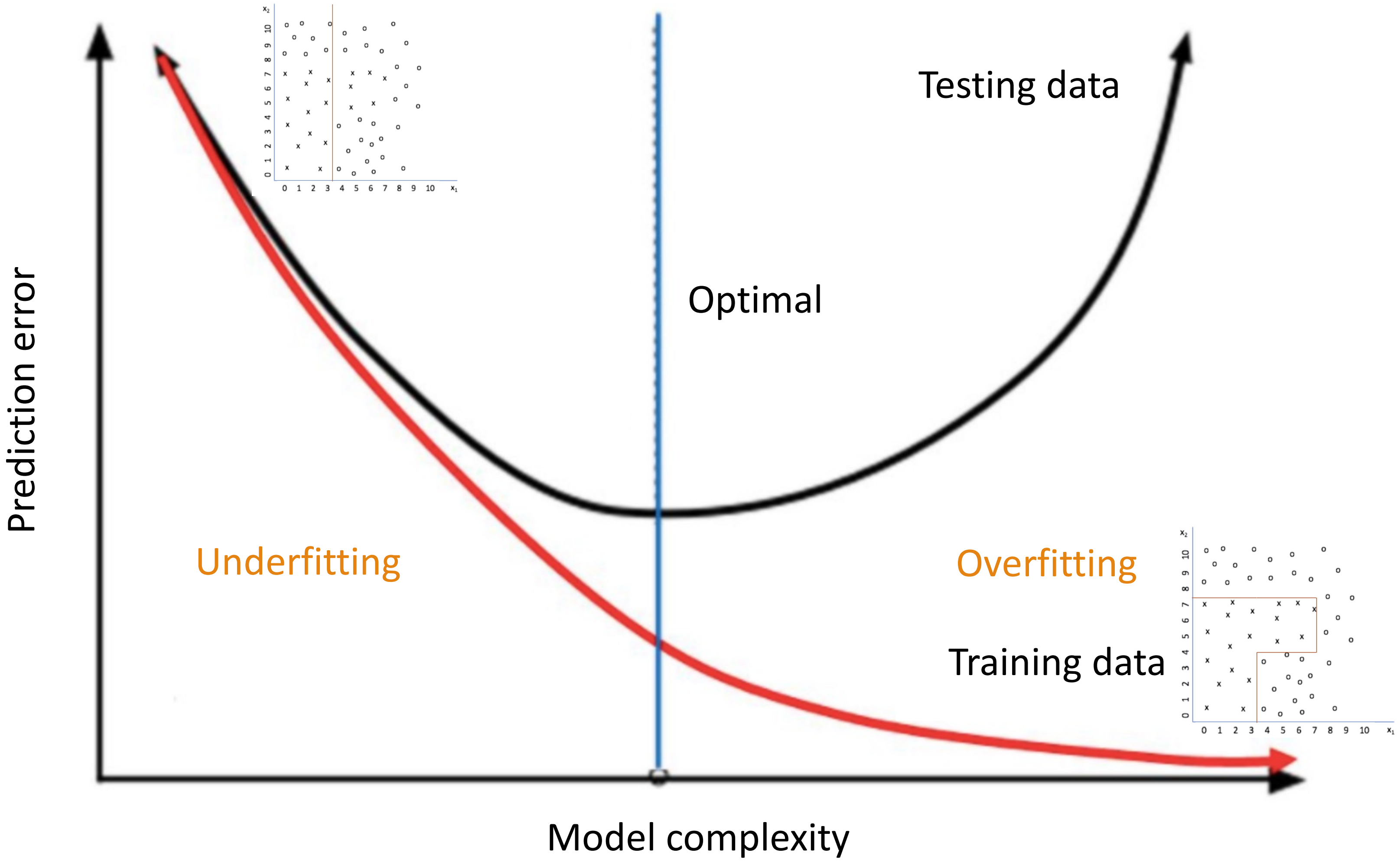
Generalization and Overfitting

- Decision Tree





Generalization and Overfitting



Generalization and Overfitting

Causes for Overfitting:

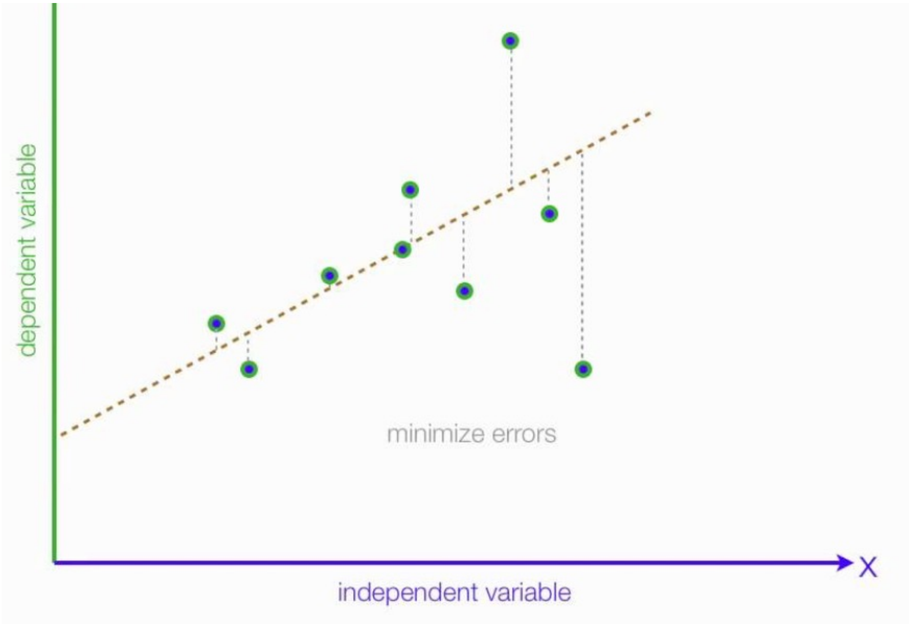
- Training data is not a good representation of testing (new) data
  - Insufficient training data
  - Noises in data: inconsistent class labels for the same values in feature set (input attributes)
  - Outliers in data: the number of samples with a given combination of class labels and feature values is small.
- An algorithm's inability to avoid overfitting noises/outliers or to train generalizable models via small amounts of training data
  - Complex model

Machine Learning Basics – Numeric Prediction

What is Numeric Prediction

- **Numeric Prediction**
  - A numeric prediction problem is when the output variable is a real value, such as “dollars” or “weight”.

- If a company’s sales have increased steadily every month for the past few years, by conducting a linear analysis on the sales data with monthly sales, the company could forecast sales in future months.



Predicting Medical Expenses

- Predicting medical expenses using linear regression
  - In order for a health insurance company to make money, it needs to collect more in yearly premium than it spends on medical care to its beneficiaries.
  - Insurance companies develop models that can accurately forecast medical expenses for its clients.
  - Goal: use patient data to estimate the medical care expenses given the health condition of a patient.
  - The estimate can be used to set the price of yearly premiums.

Simple Linear Regression

- Insurance data

	age	sex	bmi	children	smoker	region	expenses
0	19	female	27.9	0	yes	southwest	16884.92
1	18	male	33.8	1	no	southeast	1725.55
2	28	male	33.0	3	no	southeast	4449.46
3	33	male	22.7	0	no	northwest	21984.47
4	32	male	28.9	0	no	northwest	3866.86
...	...	...	...	...	...	...	...
1333	50	male	31.0	3	no	northwest	10600.55
1334	18	female	31.9	0	no	northeast	2205.98
1335	18	female	36.9	0	no	southeast	1629.83
1336	21	female	25.8	0	no	southwest	2007.95
1337	61	female	29.1	0	yes	northwest	29141.36

1338 rows × 7 columns

Simple Linear Regression

- Insurance data

	age	sex	bmi	children	smoker	region	expenses
0	19	female	27.9	0	yes	southwest	16884.92
1	18	male	33.8	1	no	southeast	1725.55
2	28	male	33.0	3	no	southeast	4449.46
3	33	male	22.7	0	no	northwest	21984.47
4	32	male	28.9	0	no	northwest	3866.86
...	...	...	...	...	...	...	...
1333	50	male	31.0	3	no	northwest	10600.55
1334	18	female	31.9	0	no	northeast	2205.98
1335	18	female	36.9	0	no	southeast	1629.83
1336	21	female	25.8	0	no	southwest	2007.95
1337	61	female	29.1	0	yes	northwest	29141.36

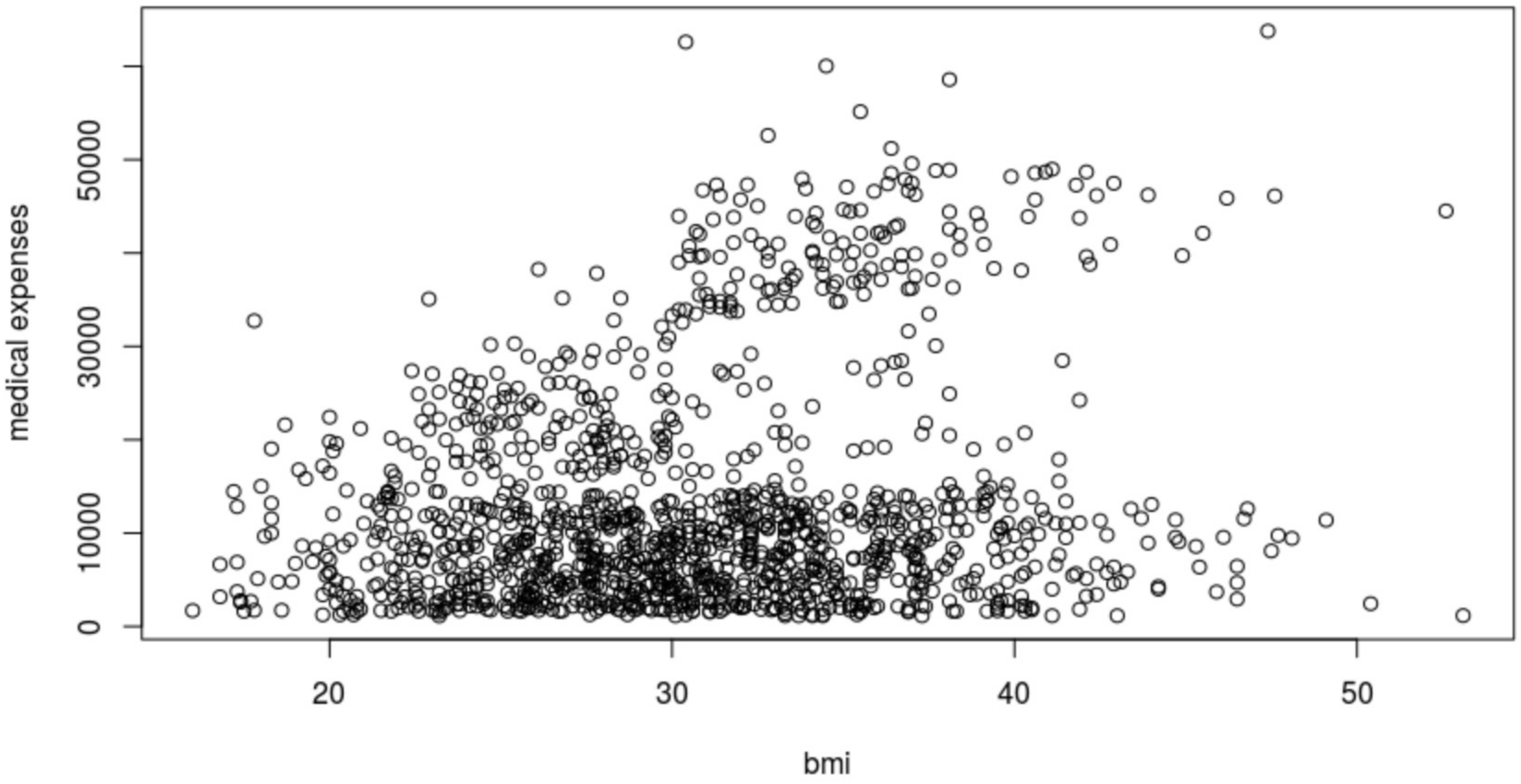
1338 rows × 7 columns

Simple Linear Regression

- Simple Linear Regression
  - Dependent variable: expenses
  - Independent variable: bmi

Simple Linear Regression

- Scatterplot of bmi and expenses



Simple Linear Regression

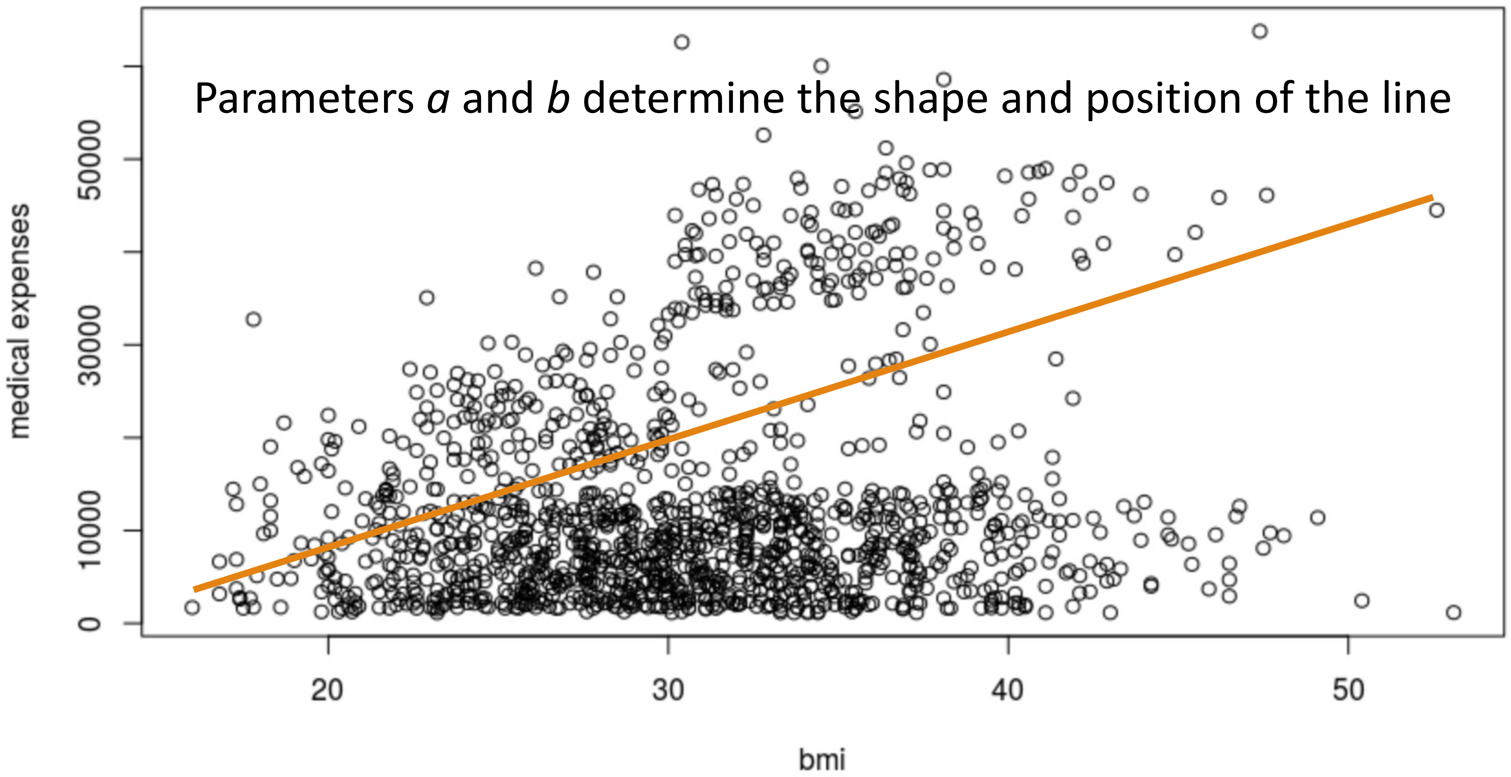
- To model the relationship between bmi and expenses, we can turn to simple linear regression.
- A simple linear regression model defines the relationship between a dependent variable and a single independent variable using a line defined by an equation in the following form:

$$y = a + bx$$

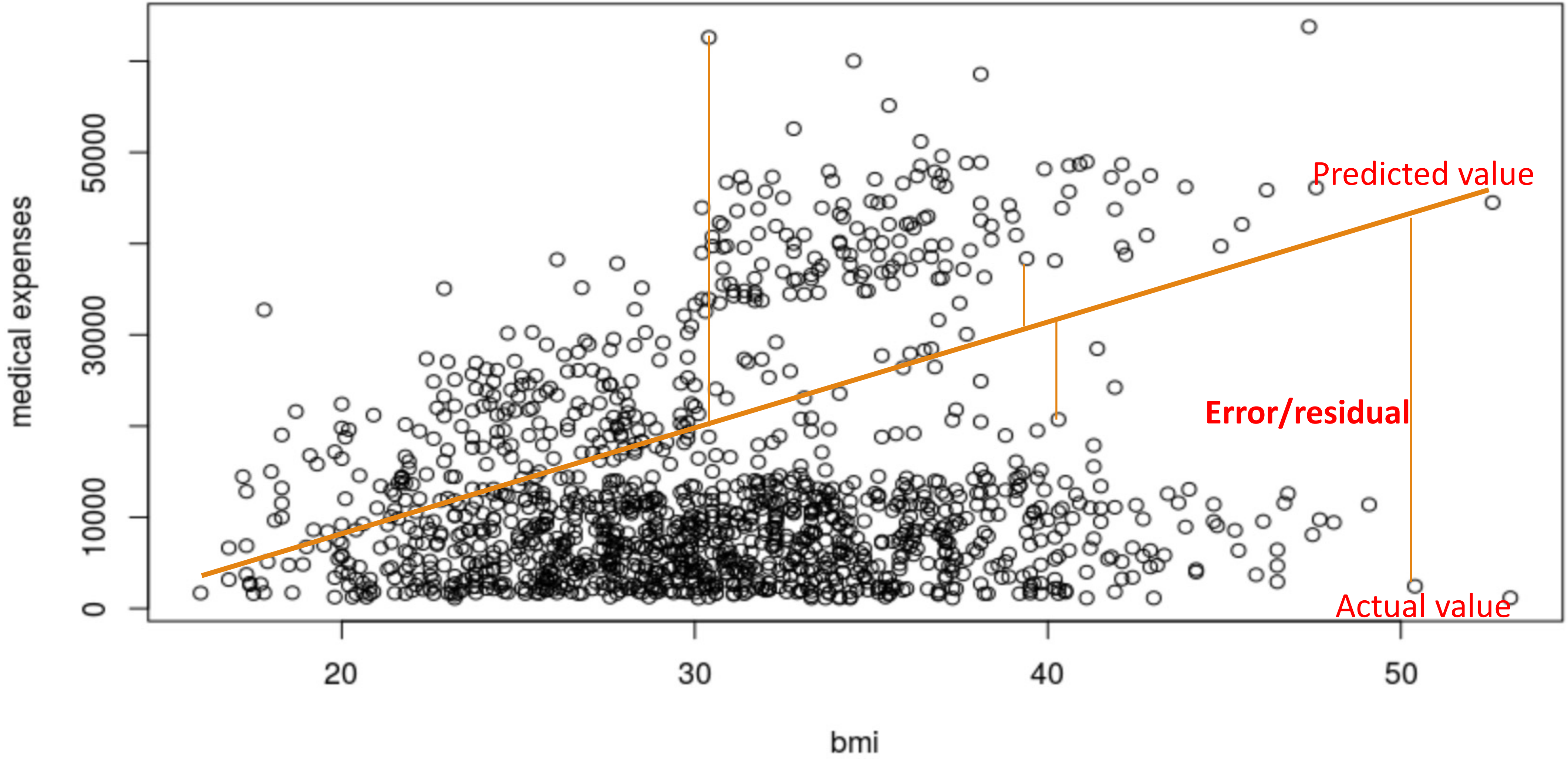
Simple Linear Regression

- Assume: expenses = 785.25 + 409.90 * bmi

The line doesn't pass through each data point exactly. Instead, it cuts through the data somewhat evenly, with some predictions lower or higher than the line.



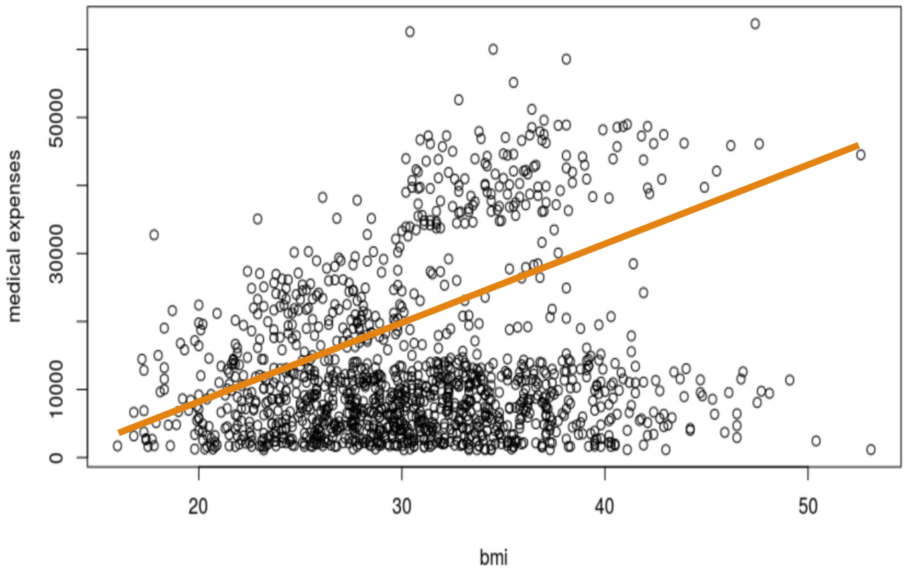
Ordinary Least Squares (OLS) Estimation



Ordinary Least Squares (OLS) Estimation

- In order to determine the optimal estimates of *a* and *b*, an estimation method known as **Ordinary Least Squares** (**OLS**) was used.
  - Minimize the residuals: $\sum (y_i - \hat{y}_i)^2 = \sum e_i^2$
    - $a = \bar{y} - b\bar{x}$ $\qquad b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$

age	sex	bmi	children	smoker	region	expenses
19	female	27.9	0	yes	southwest	16884.92
18	male	33.8	1	no	southeast	1725.55
28	male	33	3	no	southeast	4449.46
33	male	22.7	0	no	northwest	21984.47
32	male	28.9	0	no	northwest	3866.86
31	female	25.7	0	no	southeast	3756.62
46	female	33.4	1	no	southeast	8240.59
37	female	27.7	3	no	northwest	7281.51
37	male	29.8	2	no	northeast	6406.41
60	female	25.8	0	no	northwest	28923.14
25	male	26.2	0	no	northeast	2721.32



Multiple Linear Regression

- Multiple linear regression is an extension of simple linear regression.
  - $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_i x_i + \varepsilon$
  - It has additional terms for multiple independent variables
  - The dependent variable y is specified as the sum of an intercept term β_0 plus the product of the estimated β value and the x values for each of the i features.
- The goal: find values of **beta coefficients** that minimize the prediction error of a linear equation.

Multiple Linear Regression

- Step 1: Prepare data for prediction

	age	sex	bmi	children	smoker	region	expenses
0	19	female	27.9	0	yes	southwest	16884.92
1	18	male	33.8	1	no	southeast	1725.55
2	28	male	33.0	3	no	southeast	4449.46
3	33	male	22.7	0	no	northwest	21984.47
4	32	male	28.9	0	no	northwest	3866.86
...	...	...	...	...	...	...	...
1333	50	male	31.0	3	no	northwest	10600.55
1334	18	female	31.9	0	no	northeast	2205.98
1335	18	female	36.9	0	no	southeast	1629.83
1336	21	female	25.8	0	no	southwest	2007.95
1337	61	female	29.1	0	yes	northwest	29141.36

1338 rows × 7 columns

Dummy Coding

- Convert categorical variables into numeric format – **dummy coding**
  - **Create dummy variables to replace the original categorical variable.**
  - Dummy coding for a binary variable: gender

$$\text{male} = \begin{cases} 1 & \text{if x} = \text{male} \\ 0 & \text{otherwise} \end{cases}$$

Color: Red, Pink, Blue

	Color
Diamond1	Red
Diamond2	Pink
Diamond3	Blue

n categories, *n-1* dummy variables

	Color_Red	Color_Pink
Diamond1	1	0
Diamond2	0	1
Diamond3	0	0

Multiple Linear Regression

- Step2: splitting data to training and testing set

	age	bmi	children	expenses	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.9	0	16884.92	0	1	0	0	1
1	18	33.8	1	1725.55	1	0	0	1	0
2	28	33.0	3	4449.46	1	0	0	1	0
3	33	22.7	0	21984.47	1	0	1	0	0
4	32	28.9	0	3866.86	1	0	1	0	0
...	...	...	...	...	...	...	...	...	...
931	39	32.5	1	6238.30	0	0	0	0	1
932	46	25.8	5	10096.97	1	0	0	0	1
933	45	35.3	0	7348.14	0	0	0	0	1
934	32	37.2	2	4673.39	1	0	0	1	0
...	...	...	...	...	...	...	...	...	...
1333	50	31.0	3	10600.55	1	0	1	0	0
1334	18	31.9	0	2205.98	0	0	0	0	0
1335	18	36.9	0	1629.83	0	0	0	1	0
1336	21	25.8	0	2007.95	0	0	0	0	1
1337	61	29.1	0	29141.36	0	1	1	0	0

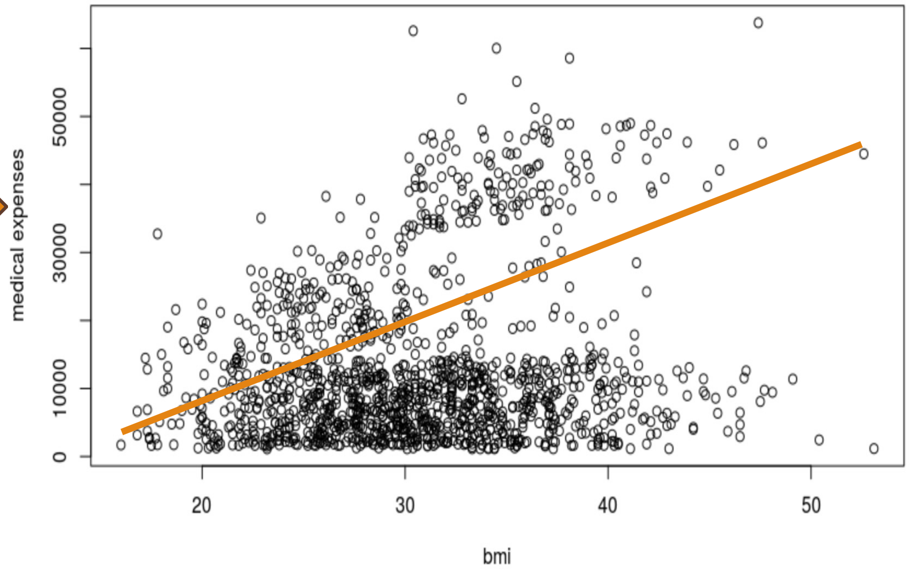
70% training data

30% testing data

Multiple Linear Regression

- Step3: Train Multiple Linear Regression model on **training data (70%)**

	age	bmi	children	expenses	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.9	0	16884.92	0	1	0	0	1
1	18	33.8	1	1725.55	1	0	0	1	0
2	28	33.0	3	4449.46	1	0	0	1	0
3	33	22.7	0	21984.47	1	0	1	0	0
4	32	28.9	0	3866.86	1	0	1	0	0
...	...	...	...	...	...	...	...	...	...
931	39	32.5	1	6238.30	0	0	0	0	1
932	46	25.8	5	10096.97	1	0	0	0	1
933	45	35.3	0	7348.14	0	0	0	0	1
934	32	37.2	2	4673.39	1	0	0	1	0
935	59	27.5	0	12233.83	0	0	0	0	1



$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_i x_i + \varepsilon$$

Calculate the coefficients

Multiple Linear Regression

- 1. The **residuals** section provides summary statistics for the errors in our predictions.
  - **Residual = real value – predicted value**
  - 50 percent of errors fall within the 1Q and 3Q values (the first and third quartile), so the majority of predictions were between $2,988.446 over the true value and $1,343.2315 under the true value.

1

```
Residuals:
        Min           1Q      Median          3Q         Max
-30302.4407  -1343.2315    985.6509    2988.446  11148.5211
```

2

```
Coefficients:
                        Estimate    Std. Error   t value    p value
_intercept          -11836.117831  1191.289921   -9.9355   0.000000
age                    256.402924    10.186903   25.1699   0.000000
bmi                    335.414456    16.372545   20.4864   0.000000
children               472.939931   166.843442    2.8346   0.004687
sex_male               -47.692891   400.980436   -0.1189   0.905348
smoker_yes           23435.046128   501.984750   46.6848   0.000000
region_northwest      -561.353117   545.419326   -1.0292   0.303645
region_southeast      -995.537125   551.901974   -1.8038   0.071580
region_southwest      -798.824371   541.710520   -1.4746   0.140648
---
```

3

```
R-squared:  0.73096,    Adjusted R-squared:  0.72864
F-statistic: 314.82 on 8 features
```

Multiple Linear Regression

- 2. The beta coefficients indicate the estimated increase in expenses for an increase of one in each of the variables, assuming all other values are held constant.
  - Intercept is of little value alone because it is impossible to have values of zero for all features

Expenses = -11836.117831 + 256.402924*age + 335.414456*bmi + 472.939931*children - 47.692891*sexmale + 23435.046128*smokeryes -561.353117*regionnorthwest – 995.537125* regionsouthest – 798.824371* regionsouthwest

1

```
Residuals:
         Min          1Q      Median          3Q         Max
-30302.4407  -1343.2315    985.6509   2988.446  11148.5211
```

2

```
Coefficients:
                        Estimate   Std. Error   t value    p value
_intercept         -11836.117831  1191.289921   -9.9355   0.000000
age                   256.402924    10.186903   25.1699   0.000000
bmi                   335.414456    16.372545   20.4864   0.000000
children              472.939931   166.843442    2.8346   0.004687
sex_male              -47.692891   400.980436   -0.1189   0.905348
smoker_yes          23435.046128   501.984750   46.6848   0.000000
region_northwest     -561.353117   545.419326   -1.0292   0.303645
region_southeast     -995.537125   551.901974   -1.8038   0.071580
region_southwest     -798.824371   541.710520   -1.4746   0.140648
---
```

3

```
R-squared:  0.73096,    Adjusted R-squared:  0.72864
F-statistic: 314.82 on 8 features
```

Multiple Linear Regression

- 3. The **multiple R-squared value** provides a measure of how well our model as a whole explains the values of the dependent variable.
  - The model explains nearly 73 percent of the variation in the dependent variable.
  - High R-squared on training data indicates overfitting.

1

```
Residuals:
         Min           1Q      Median          3Q         Max
-30302.4407 -1343.2315  985.6509  2988.446  11148.5211
```

2

```
Coefficients:
                        Estimate    Std. Error  t value   p value
_intercept          -11836.117831  1191.289921  -9.9355  0.000000
age                    256.402924    10.186903  25.1699  0.000000
bmi                    335.414456    16.372545  20.4864  0.000000
children               472.939931   166.843442   2.8346  0.004687
sex_male               -47.692891   400.980436  -0.1189  0.905348
smoker_yes           23435.046128   501.984750  46.6848  0.000000
region_northwest      -561.353117   545.419326  -1.0292  0.303645
region_southeast      -995.537125   551.901974  -1.8038  0.071580
region_southwest      -798.824371   541.710520  -1.4746  0.140648
---
```

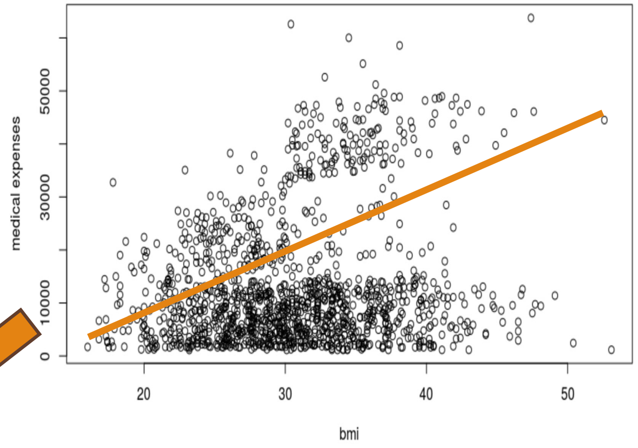
3

```
R-squared:  0.73096,     Adjusted R-squared:  0.72864
F-statistic: 314.82 on 8 features
```

Multiple Linear Regression

- Step 4: make predictions on **testing data (30%) data.**

	age	bmi	children	expenses	sex_male	smoker_yes	region_northwest	region_southeast	region_southwest
0	19	27.9	0	16884.92	0	1	0	0	1
1	18	33.8	1	1725.55	1	0	0	1	0
2	28	33.0	3	4449.46	1	0	0	1	0
3	33	22.7	0	21984.47	1	0	1	0	0
4	32	28.9	0	3866.86	1	0	1	0	0
...	...	...	...	...	...	...	...	...	...
931	39	32.5	1	6238.30	0	0	0	0	1
932	46	25.8	5	10096.97	1	0	0	0	1
933	45	35.3	0	7348.14	0	0	0	0	1
934	32	37.2	2	4673.39	1	0	0	1	0
...	...	...	...	...	...	...	...	...	
1333	50	31.0	3		1	0	1	0	0
1334	18	31.9	0		0	0	0	0	0
1335	18	36.9	0		0	0	0	1	0
1336	21	25.8	0		0	0	0	0	1
1337	61	29.1	0		0	1	1	0	0



Multiple Linear Regression

- Step 4: make predictions on **testing data (30%) data**

expenses	Predicted expenses
16884.92	15445.54
1725.55	1455.21
4449.46	6799.98
21984.47	12351.13
3866.86	2434.12
3756.62	1235.12
8240.59	6394.12
7281.51	5450.74
6406.41	2509.23
28923.14	39455.26
2721.32	2311.65

→ Examine prediction performance

Evaluation Metrics for Numeric Prediction

- **Mean Absolute Error (MAE)**
  - The **mean absolute error (MAE)** is a quantity used to measure how close forecasts or predictions are to the eventual outcomes.
  - This is known as a scale-dependent accuracy measure and therefore cannot be used to make comparisons between series using different scales

$$\text{MAE} = \frac{1}{\text{n}}\sum_{t=1}^{n} |y_t - \hat{y}_t|$$

- **Root Mean Squared Error (RMSE)**
  - **RMSE** gives a relatively high weight to large errors. This means the RMSE should be more useful when large errors are particularly undesirable.

$$\text{RMSE} = \sqrt{\frac{\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}{n}}$$

```
MAE: 4011.3808850910846
RMSE: 5773.7111375700415
```

Numeric Prediction vs Classification

- **Numerical Prediction**
  - A numerical prediction problem is when the output variable is a real value, such as “dollars” or “weight”.